

Homework 11 Solutions

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Problem 24.1

- a) 1) Yes
2) $\{1, 3\}, \{2, 4\}$
3) Yes
4) $\{(2, 1), (4, 3)\}$
- b) 1) Yes
2) $\mathbb{Z}$, all even integers
3) Yes
4) $\{(x, y) : x, y \in \mathbb{Z}, x = 2y\}$
- c) 1) Yes
2) $\mathbb{Z}, \mathbb{Z}$
3) Yes
4) $\{(x, y) : x, y \in \mathbb{Z}, y + x = 0\}$
- d) 1) No, $(0, 1) \in f$ and $(0, 2) \in f$.
- e) 1) Yes
2) $\mathbb{Z}, \{y : x, y \in \mathbb{Z}, y = x^2\}$
3) No, $f(2) = 4$ and $f(-2) = 4$
- f) 1) Yes
2) $\emptyset, \emptyset$
3) Yes
4) $\emptyset$
- g) 1) No, $(0, 1) \in f$ and $(0, -1) \in f$
- h) 1) No, $(1, 2) \in f$ and $(1, 3) \in f$
- i) 1) Yes
2) $\mathbb{N}, \mathbb{N}$
3) Yes
4) $\{(x, x) : x \in \mathbb{N}\}$
- j) 1) No, $(5, 0) \in f$ and $(5, 5) \in f$

Problem 24.3

neither $\{(1, 3), (2, 3)\}, \{(1, 4), (2, 4)\}, \{(1, 5), (2, 5)\}$

one-to-one $\{(1, 3), (2, 4)\}, \{(1, 3), (2, 5)\}, \{(1, 4), (2, 3)\}, \{(1, 4), (2, 5)\}, \{(1, 5), (2, 3)\}, \{(1, 5), (2, 4)\}$

Problem 24.5

- a) -2
- b) 3
- c) 27
- d) 1
- e) 2

Problem 24.6

- a) $\mathbb{Z}$
- b) $\{y \in \mathbb{Z} : y \geq 0\}$
- c) $\mathbb{Z}$
- d) $(0, 1]$
- e) $\{y \in \mathbb{Z} : y \geq 0\}$
- f) $[0, 1]$

Problem 24.16

Let $f : A \rightarrow B$ for the following proofs.

Proof. If $f : A \rightarrow B$ is one-to-one and onto, then $|A| = |B|$.

1. Since f is a function, then $|A| = |\text{dom } f|$.
2. Since f is one-to-one, then $|\text{dom } f| = |\text{im } f|$.
3. Since f is onto, then $|\text{im } f| = |B|$.
4. Therefore $|A| = |B|$. □

Proof. If $f : A \rightarrow B$ is one-to-one and $|A| = |B|$, then f is onto.

1. Since f is a function, then $|A| = |\text{dom } f|$.
2. Since f is one-to-one, then $|\text{dom } f| = |\text{im } f|$.
3. Since $|A| = |B|$, then $|\text{im } f| = |B|$.
4. Therefore f is onto. □

Proof. If $f : A \rightarrow B$ is onto and $|A| = |B|$, then f is one-to-one.

1. Since f is a function, then $|A| = |\text{dom } f|$.
2. Since f is onto, then $|\text{im } f| = |B|$.
3. Since $|A| = |B|$, then $|\text{dom } f| = |\text{im } f|$.
4. Therefore f is one-to-one. □